**Exp-6:**
Create a simple Calculator application in Flutter.

```
import 'package:flutter/material.dart';

void main() => runApp(const CalculatorApp());

class CalculatorApp extends StatelessWidget {
 const CalculatorApp({super.key});

 @override
 Widget build(BuildContext context) {
  return MaterialApp(
    title: 'Simple Calculator',
    theme: ThemeData(
      primarySwatch: Colors.deepPurple,
      scaffoldBackgroundColor: Colors.white,
      inputDecorationTheme: const InputDecorationTheme(
        border: OutlineInputBorder(),
      ),
      elevatedButtonTheme: ElevatedButtonThemeData(
        style: ElevatedButton.styleFrom(
          padding: const EdgeInsets.symmetric(vertical: 16),
          textStyle: const TextStyle(fontSize: 20),
        ),
      ),
    ),
    home: const CalculatorScreen(),
  );
 }
}

class CalculatorScreen extends StatefulWidget {
 const CalculatorScreen({super.key});

 @override
 State<CalculatorScreen> createState() => _CalculatorScreenState();
}

class _CalculatorScreenState extends State<CalculatorScreen> {
 final TextEditingController _controller1 = TextEditingController();
 final TextEditingController _controller2 = TextEditingController();
 String _result = '';

 double? _parseInput(String input) {
  return double.tryParse(input.trim());
 }
```

```
void _calculate(String operation) {
  final num1 = _parseInput(_controller1.text);
  final num2 = _parseInput(_controller2.text);

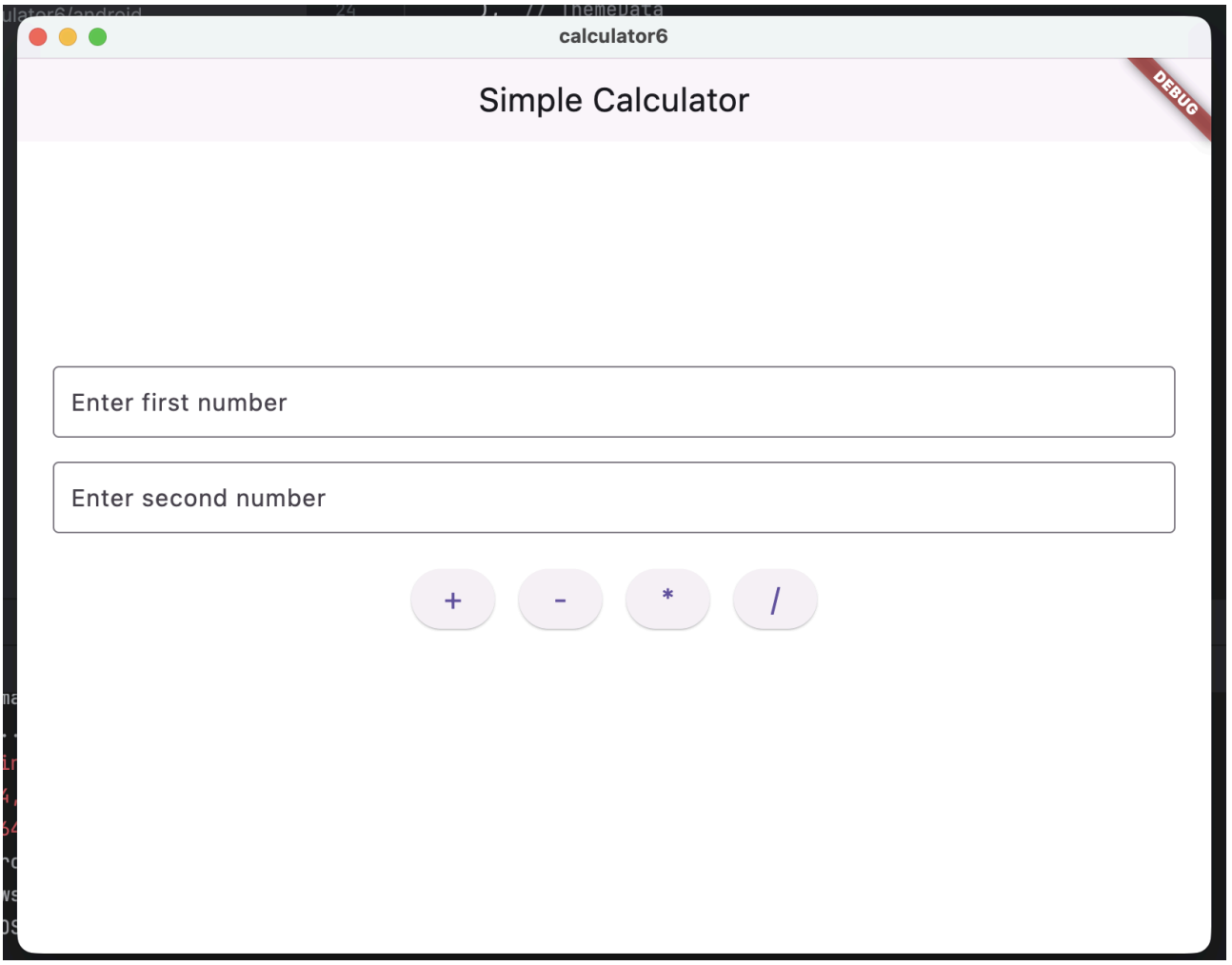
  if (num1 == null || num2 == null) {
    setState(() {
      _result = 'Please enter valid numbers';
    });
    return;
  }

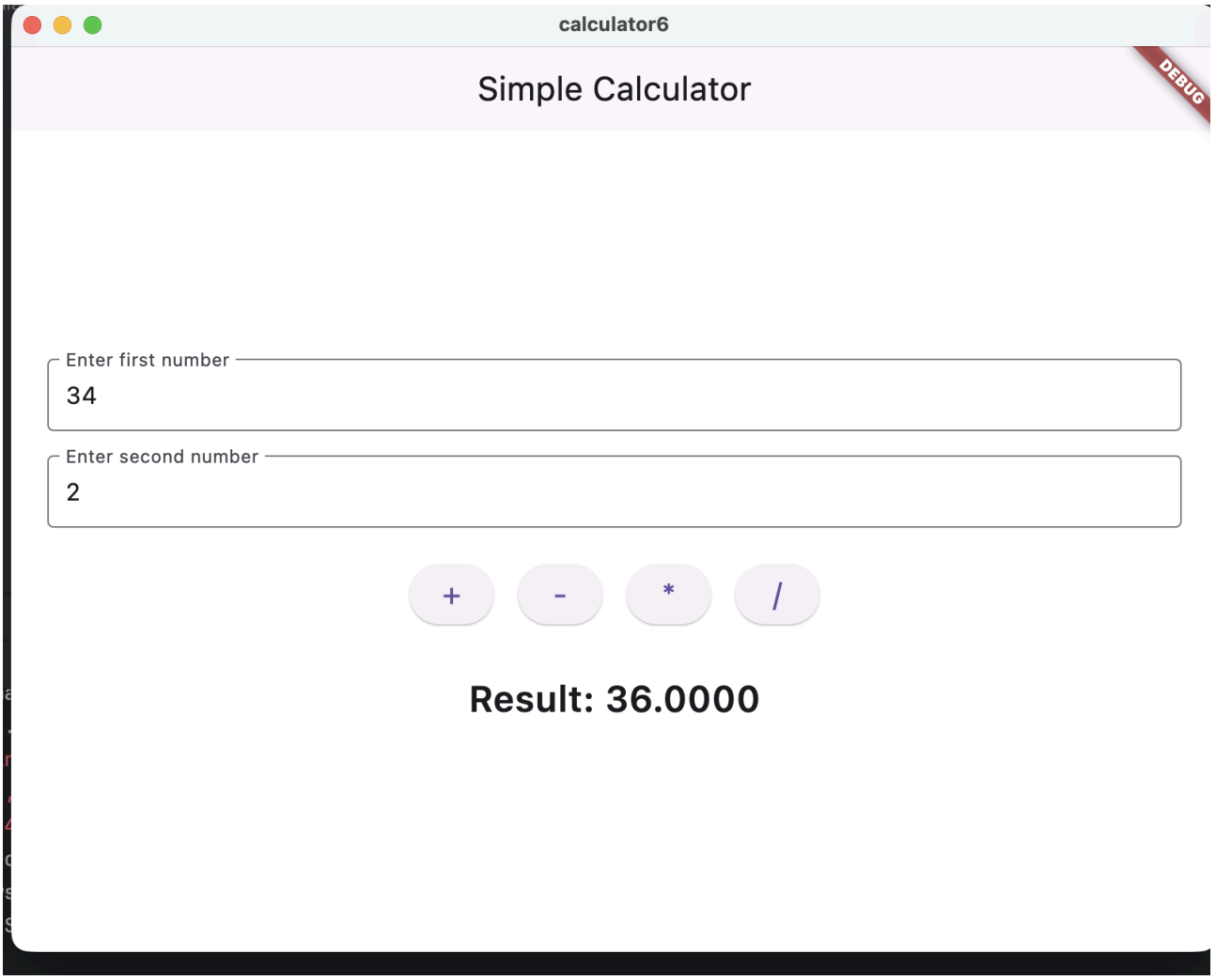
  double res;
  switch (operation) {
    case '+':
      res = num1 + num2;
      break;
    case '-':
      res = num1 - num2;
      break;
    case '*':
      res = num1 * num2;
      break;
    case '/':
      if (num2 == 0) {
        setState(() {
          _result = 'Cannot divide by zero';
        });
        return;
      }
      res = num1 / num2;
      break;
    default:
      res = 0;
  }
  setState(() {
    _result = 'Result: ${res.toStringAsFixed(4)}';
  });
}

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(title: const Text('Simple Calculator')),
    body: Padding(
      padding: const EdgeInsets.all(24),
      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [
```

```
          TextField(
            controller: _controller1,
            keyboardType: const TextInputType.numberWithOptions(decimal: true),
            decoration: const InputDecoration(
              labelText: 'Enter first number',
            ),
          ),
          const SizedBox(height: 16),
          TextField(
            controller: _controller2,
            keyboardType: const TextInputType.numberWithOptions(decimal: true),
            decoration: const InputDecoration(
              labelText: 'Enter second number',
            ),
          ),
          const SizedBox(height: 24),
          Wrap(
            spacing: 16,
            children: [
              ElevatedButton(
                onPressed: () => _calculate('+'),
                child: const Text('+'),
              ),
              ElevatedButton(
                onPressed: () => _calculate('-'),
                child: const Text('-'),
              ),
              ElevatedButton(
                onPressed: () => _calculate('*'),
                child: const Text('*'),
              ),
              ElevatedButton(
                onPressed: () => _calculate('/'),
                child: const Text('/'),
              ),
            ],
          ),
          const SizedBox(height: 32),
          Text(
            _result,
            style: const TextStyle(fontSize: 24, fontWeight: FontWeight.w600),
          ),
        ],
      ),
    ),
  );
 }
}
```

**Output:**


calculator6
Simple Calculator
DEBUG
Enter first number
Enter second number
+
-
*
/


calculator6
Simple Calculator
DEBUG
Enter first number
34
Enter second number
2
+
-
*
/
Result: 36.0000